REPLICATION REPORT

Schoonvelde, M., Brosius, A., Schumacher, G., Bakker, B. (2019). *Liberals lecture, conservatives communicate: Analyzing complexity and ideology in 381,609 political speeches*. PLOS |ONE.

Replication GitHub located here

EXECUTIVE SUMMARY

My replication of Schoonvelde et al. (2019) lends support to the validity of their findings. However, my replication also raises concerns about the authors' calculation of Flesch-Kincaid (FK) Grade scores. The authors do not clearly explain or show in their code how they count words, sentences and syllables to calculate the FK scores, or how any pre-processing of tokens may have impacted their counts. There is variation between the FK scores generated in my replication and the authors' FK scores. Not only should the linguistic complexity calculation method be more precise, but precision may be a struggle without human validation of languages.

INTRODUCTION

The authors seek to further establish a causal link between political party ideology and linguistic complexity. Limited empirical evidence in the United States and the United Kingdom lends support to the hypothesis that liberals speak differently than conservatives. Liberals – due to their more "open minded" nature with regards to policy issues and their greater tolerance for ambiguity – are theorized to speak in more lengthy and complex ways than conservatives. Schoonvelde et al. broaden the context of study to Europe (UK, Spain, Netherlands, Denmark, Germany and Sweden) and to more diverse audiences (parliamentary speeches, EU speeches and internal party congress speeches). The authors find support for the following: a) more "culturally liberal" political parties use more complex language, b) there is mostly no impact on linguistic complexity due to "left/right" economic ideology, c) there is a mixed impact of being the "governing party" on complexity, d) there is a downward time trend of linguistic complexity.

DATA & METHODS: AUTHORS

Sample

The authors' sample consists of 381,609 speeches, which represent 10 countries, span 70 years (1945-2015) and contain 12 European prime ministers. The overall sample is found in the *speeches.RData* in the GitHub. The authors draw upon 3 distinct datasets and 8 corpora, which are also found in the GitHub (*parlspeech.RData, euspeech.RData and congressspeech.RData*). The Parliamentary Speeches sub-set contains 385,275 speeches in 5 parliaments (British House of Commons, Germany's Bundestag, Netherland's Tweede Kamer, Spain's Congresso and Sweden's Riksdag); these speeches are not translated. The EU Speeches sub-set contains 1,847 speeches given by heads of government at the EU, featuring leaders from 10 different European countries; these speeches were spoken in English at the time at the EU. Lastly, the Party Congress sub-set contains 1,187 speeches from Denmark and the Netherlands which were given at congress meetings across 12 total political parties; these speeches are not translated.

Unit of Analysis and Pre-Processing

The authors' unit of analysis is the entire speech. Notably, only speeches with more than 10 sentences (at least 11) containing 5+ words were included in the sample. The entire possible sample of parliament, EU and congress speeches before this cutoff of 11 sentences with 5+

words is not available for analysis. The authors appear to engage in some slight pre-processing of tokens within their selected samples. As shown in the validation datasets in the GitHub (e.g., *HouseofCommons_Sample.Rdata*) the authors sometimes reduce a handful of words from each speech. The logic behind this pre-processing is not stated.

Dependent Variable: Linguistic Complexity

The authors use the Flesch-Kincaid Grade score (FK) to calculate linguistic complexity. FK has been primarily rooted in English language analysis and used in an educational setting to assess the grade-level of teaching material for students. FK is based on word count, sentence count and syllable count to assess complexity. Complexity becomes a weighted average of sentence length and word length, represented by this formula. While FK can be calculated within R using the Quanteda package (textstat_readability), it is unclear from the authors' code if they used R to generate their FK scores or used other computational techniques to calculate word count, sentence count and syllable count and then applied the formula. The FK scores are pre-calculated and appear as is in all datasets and sample datasets.

Flesch–Kincaid Grade Level

$$0.39\left(\frac{\text{total words}}{\text{total sentences}}\right)+11.8\left(\frac{\text{total syllables}}{\text{total words}}\right)-15.59$$

Independent Variables: Political Ideology, Time and Governing Party

The authors rely upon an independent organization, called the Manifesto Project, to calculate two separate measures of political ideology. The Manifesto Project measures the yearly election platforms of global political parties. Based on the pro/anti stance of the platform across 53 issues, the Manifesto Project generates a left/right or liberal/conservative ideology score. The authors use the Manifesto Project to create two separate ideology splits: liberal/conservative on social issues and economic issues. To calculate each, the authors sum and log transform the more liberal leaning issues and then subtract the log transformed sum of the conservative leaning issues. They let the ideology score for the election platform one year earlier (t-1) represent the ideology of the party and speaker in present day (t). However, the replication process is not found in the authors' code; the ideology scores are found in all datasets pre-calculated. The authors also use speech year and membership in the governing party as independent variables.

Fixed Effects OLS Regression

The authors use fixed effects on speaker in some of their OLS regression analysis. The authors complete 8 OLS regressions, one for each corpora (5 parliaments, 1 EU, 2 congresses).

REPLICATION APPROACH

I located the replication files provided by the authors. This Harvard Dataverse address is located here. I chose the replication approach below, which drew upon the following files:

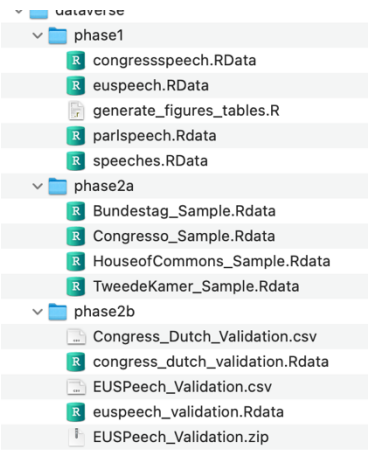
Phase 1: Replicate from Pre-Processed Data

The large datasets provided (speeches, parlspeech, euspeech and congressspeech) are pre-processed; linguistic complexity and political ideology scores are given but not calculated. The authors' *generate_figures_tables.R* script provides all the code needed to generate the key tables and figures in the report and the appendix. I wanted to use the generate_figures_table R code as is to test that the datasets provided generate identical results – as they should. I used the

speeches, parlspeech, euspeech and congressspeech datsets from R, depending on what the generate_figures_table code required. This process is documented in Sara's Replication – Phase 1.RMarkdown on GitHub.

Phase 2A: Replicate from Unprocessed Data (Parliament)

To complete a true replication paper, however, I needed to replicate the FK scores. By reviewing the "validation" datasets found in Dataverse, I discovered that 4 of the 5 parliament corpora were provided that contain raw speech text data, along with a calculated FK score, word count, sentence count and syllable count. However, this sample was only a small portion (roughly 16,000 speeches or 4% of total Parliamentary speeches). I generated my own FK score, and word/sentence/syllable counts and compared these results to the authors' work. I used the R Quanteda package (textstat_readability) to generate the FK score and parcel out the underlying word, sentence and syllable counts. I then re-ran the same generate_figures_table.R script used in the Phase 1 RMarkdown to generate key figures and tables using the new replicated FK scores. This process is documented in Sara's Replication – Phase 2a 2b.RMarkdown on GitHub.



Phase 2B: Replicate from Unprocessed Data (EU and Congress Netherlands)

The full corpora for EU speeches and half of the congress corpora (Netherlands only) were also available unprocessed with raw speech data. Once again, I decided to use generate my own calculations of FK, word/sentence/syllable count and compare the replicated complexity results to the authors' work. I then re-ran the same script used in the Phase 1 RMarkdown to generate key figures and tables using the replicated FK scores. This process is documented in Sara's Replication – Phase 2a 2b.RMarkdown on GitHub.

I ultimately decided due to time constraints and concerns about replication validity to not replicate the ideology scores calculated from the Manifesto Project. As a result, how accurate the ideology measures are is not answerable from my replication process. The authors did not provide the full, unprocessed dataset of all speeches on Dataverse. However, it is possible to manually reconstruct these datasets using the footnotes in the paper. I chose not to do so for time constraints and concerns about replication validity. I preferred to work with the sample validation data provided – where the authors' FK score, word/sentence/syllable count was stated alongside raw speech text – to generate meaningful replication comparisons.

RESULTS AND KEY DIFFERENCES

Replication GitHub contains presentation slides with visuals displaying replication results

Phase 1: The Phase 1 replication mostly replicated identically – as it should, given it was the exact dataset and code supplied by the authors. However, as shown in the presentation slides, I saw some variation in **Figure 1** - which measured mean linguistic complexity across select Parliamentary speakers. With respect to the two Spanish speakers, in my replicated results, the socially conservative candidate (Rajoy) was more complex than the socially liberal candidate (Zapatero); this was the opposite of the authors' findings. Additionally, the mean complexity for

both speakers in my replicated results were around 20, whereas the authors' results for both speakers were slightly below 15. Upon closer investigation, I discovered in the parliamentary speeches dataset there are several Spanish candidates with the name Zapatero and Rajoy. My replicated figures were built by selecting on the full specific speaker name (i.e. zapatero rodriquez and rajoy brey). The authors only selected on a single last name for each candidate to build Figure 1. Given the Spanish candidates themselves sometimes have multiple family names and that there were multiple Spanish candidates with the name zapatero and rajoy, I suspect the selection of speaker names could have been more precise for Figure 1. This is a situation where the incorporation of multiple languages and cultures into a dataset makes it more challenging to generalize code effectively across the sample.

Phase 2a: The Phase 2a replication first revealed the uncertainty of how the FK scores were calculated by the authors for linguistic complexity. After calculating my FK scores, word/sentence/syllable counts using the R Quanteda package, I then developed graphs and correlation scores to compare the FK scores and word/sentence/syllable counts between the replication and the authors' work. As shown graphically in the presentation slides, for all 4 countries, there was variation between mine and the authors' FK scores: UK, 85.5% correlation; Spain, 95% correlation; Germany, 97.5% correlation; Netherlands, 98.6% correlation. Upon closer inspection, the sentence count and syllable counts appear to be the culprit of the discrepancies. The average correlation between replicated and author sentence count and syllable count was typically between 97-99%. However, given the calculation of FK, even small discrepancies in syllable counts can have large impacts on FK scores. What is most interesting about these results is the fact the FK score was originally designed for English based text but the English parliamentary speeches have the lowest FK correlation (replicated vs authors).

I then replicated the figures and tables using the sample parliament dataset with my FK scores as the complexity measure. It is important to reiterate that this sample is only 4% of the total parliamentary speeches and thus differences between the replicated and author figures/tables may stem from this issue rather than the FK scores themselves. That said, the major differences were as follows: Figure 1 once again had socially conservative Rajoy as more complex than socially liberal Zapatero in Spain, and had the mean complexity scores close to 20. This time, Figure 1 also had socially conservative Cameron as more complex than socially liberal Brown in the UK, but the mean complexity scores were at similar values as the authors' Figure 1. For Figure 2a, being socially conservative made a speaker more linguistically complex in the UK, with an average effect of approximately 0.1; the authors found a -0.25 impact of conservatism on complexity in the UK. For Figure 2b, being economically conservative had an average negative effect of -0.175 in the UK and had an average positive effect just below 0.10 in Spain (which the authors also found). For Figure 3, while linguistic complexity continued to decrease over time in Spain, the Spanish average complexity hovered around 20, which was substantially higher than the authors' findings that hovered around 12. The UK's complexity scores in the 2010s began to lower to 1990s levels, whereas they rise sharply after the 1990s in the authors' Figure 3.

Phase 2b: I found similar results when comparing my replicated FK scores to the authors' FK scores. Specifically, the EU speeches had an FK correlation of 95% and the Netherlands Congress speeches had an FK correlation of 98.7%. Once again, the Dutch text was more closely

aligned with the R Quanteda calculation I made. Once again, sentence counts and syllable counts were contributing factors to these differences in FK scores.

When replicating the figures and tables, these results were much closer to the authors' figures and tables than in Phase 2a. This gives hope that – despite FK score variation – the overall results still hold. It makes sense that the Phase 2b results would be more informative from a replication standpoint given the unprocessed EU and Netherlands Congress samples represent the full sample of texts used by the authors (unlike with Phase 2a). In terms of notable differences, in Figure 2a, heads of government (i.e EU attendees) had a larger impact on complexity (approximately -0.40 rather than -0.20) in my replicated results but was still negative.

REPLICTATION AUTOPSY

What worked the best in this replication project was the script provided by the authors to generate the tables and figures. Aside from a few areas where the code was missing some features (such as in Figure 1, specifying full speaker names), the code ran very easily and was sorted into clear, definable chunks linked to specific figures and tables. The main area of the replication that did not work well – or as desired - was the replication of the FK scores for two major reasons: 1) I did not have access to the full unprocessed speech data, in particular for the 5 parliament corpora and the Denmark congress speeches, with the authors' FK scores and word/sentence/syllable counts visible and 2) it was unclear how the authors' calculated the FK score because there was no code to generate the scores included in the dataset. As a result, I took the approach I did of replicating the unprocessed corpora I had and comparing FK, word count, sentence count and syllable count both visually and through correlation scores. However, my replication would be strengthened if I was able to complete this analysis on all 380,000 speeches and knew precisely how the authors' calculated the FK score. In terms of where I got stuck, I decided – given the amount of effort needed to replicate the FK scores – that I would refrain from re-creating the ideology scores. Once again, code was missing for me to precisely recreate how the authors created ideology scores per speaker per year using the Manifesto database and I did not want to recreate this work inaccurately and hurt the validity of my results.

REPLICATION EXTENSION

In terms of what I would have done differently to increase the precision of the results (besides clearly stating how the FK and ideology scores were calculated and providing relevant code), I would have done the following: 1) I would have changed the unit of analysis to the paragraph or sentence block level to see if the results change. While some of the speeches have only 11-15 sentences, others have 75-200 sentences. It would be important to account for differences in size of speech, whereas right now each speech is treated the same regardless of its length. 2) I would want to show I have engaged in human validation to support the FK scores, in particular the word, sentence and syllable count. I would build a research team of academics or research assistants who are fluent in each of the languages represented in the sample to validate the FK scores. In terms of innovation, I would be interested in incorporating structural topic models (STM) to account for which speech topics are complexity differences most relevant, and how this complexity is impacted by political ideology as a moderator. In terms of important research questions STM would help address: 1) Topic selection: Do left vs right speakers emphasize different topics? 2) Linguistic complexity: Are those topics expressed with different complexity? 3) Interaction: Does ideology change how simply a topic is discussed?